Southern Cross University

Third Part/Chapter of the Autonomic Nervous System (ANS) Pharmacology

Introductory Pharmacology
PHAR2001

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Objectives

Describe the actions interactions and side effects of the following adrenergic antagonists:

- Phentolamine and labetalol (non-selective antagonist)
- Prazosin and doxazosin (an alpha 1 selective antagonist)
- Metoprolol and atenolol (a beta 1 selective adrenergic antagonist)

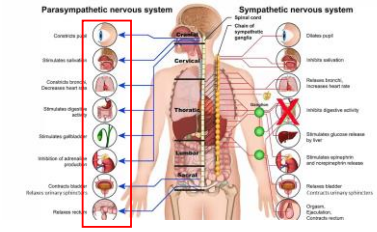
These medications will be taken again at a later time. It is most important right now to understand why the actions and side effects occur.

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Sympatholytics (drugs that block the SNS)
Lesson 1:
Understanding drugs that block the SNS

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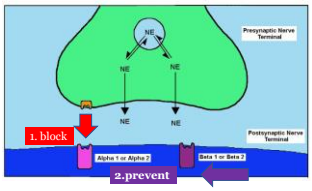
Sympatholytic agents allow the parasympathetic nervous system dominate



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Sympatholytic drugs

1. **Block** adrenoreceptors on the surface of an effector or
2. **Prevent** transmission by adrenergic nerves



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Sympatholytics
Lesson 2:
Non-selective adrenergic blockers

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Non-selective sympatholytic drugs

Mostly used to treat hypertension

Labetalol (know this drug; used to treat pre-eclampsia)

Adverse effects

- Postural hypotension
- Blurred vision
- Bronchospasm (dyspnoea; contraindicated in asthma)
- Swollen ankles
- Nasal congestion

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Pre-eclampsia can advance to life threatening eclampsia



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Alpha antagonists

Recall the location of receptors:

- Blood vessels
- Smooth muscles of the GIT
- Radial muscles of the eye
- Sphincters of the bladder

If you block the adrenergic activity of these,

- Vasodilation = decreased blood pressure
- Increased GI motility and secretions
- Pupil constriction
- Increased micturition

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Non-selective *Alpha* antagonists

Phentolamine (Regitine; know this drug)

Used to:

- Control hypertensive emergencies, cocaine overdose, and adrenaline overdose
- Locally to increase blood flow for instance in the location of an injection of adrenaline

Side effects:

- Nasal congestion
- Postural (orthostatic) hypotension

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Sympatholytics
Lesson 3:
Selective adrenergic blockers (Part 1: Alpha 1 blockers)

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Alpha 1 antagonists

Receptors in the blood vessels and prostate

Drugs end in '…**osin**' alfuz**osin**, praz**osin**…

Treatment of

- Hypertension
- Benign prostatic hyperplasia

- May result in hypotension!!

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Alpha 1 antagonists

Receptors in the blood vessels and prostate

Drugs end in '...osin' alfuzosin, prazosin...

Treatment of
- Hypertension
- Benign prostatic hyperplasia

- May result in hypotension!!

Highly selective alpha 1a drugs are preferred for prostate

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Sympatholytics
Lesson 4:
Selective adrenergic blockers
(Part 2: Beta blockers)

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Beta antagonists (...olol drugs)

Can be non-selective:

- Propranolol (know this drug; first of this class, but crosses blood brain barrier)
- Timolol (know this drug; treatment of choice in the patient with hypertension and angina)
- Acebutolol... (other examples)

Or selective for Beta 1 or Beta 2 receptors, but there are no clinical applications for Beta 2 antagonists

Selective Beta 1 antagonists were developed to reduce bronchospasm

- Contraindicated for those with asthma, COPD, allergies

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Beta 1 antagonists

The blockage of the beta 1 receptors in the heart

- Decreased chronotropic effects
- Decreased inotropic effects
- Together, these have the effect of
  - stabilising the rhythm and
  - leading to decrease in arrhythmias,
  - decreased cardiac workload
  - decreased oxygen consumption

The blockage of the beta 1 receptors in the kidneys

- Renin- angiotensin-aldosterone mechanism decreased
- Decreased blood pressure

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Beta 1 antagonists

Metoprolol (lopressor; know this drug)
Betaxolol, Atenolol

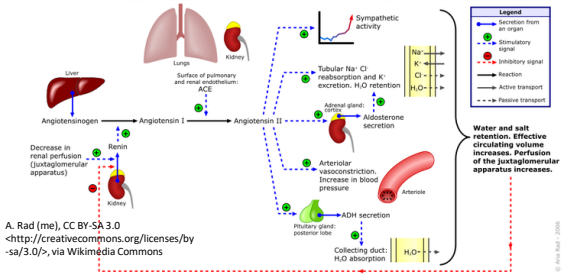
Adverse reactions:

- Dizziness
- Bradycardia
- Orthostatic hypotension
- Bronchospasm

Never stop this drug abruptly!

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The renin angiotensin aldosterone system increases water and salt retention



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There is no clinical use for Beta 2 antagonists in humans

Why?

Recall the location of the receptors:

- **Lung**
- **Mast cells**
- **Skeletal muscle**

(These would constrict the bronchioles and make mast cells more sensitive, so there is no clinical application for this type of drug)

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References

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